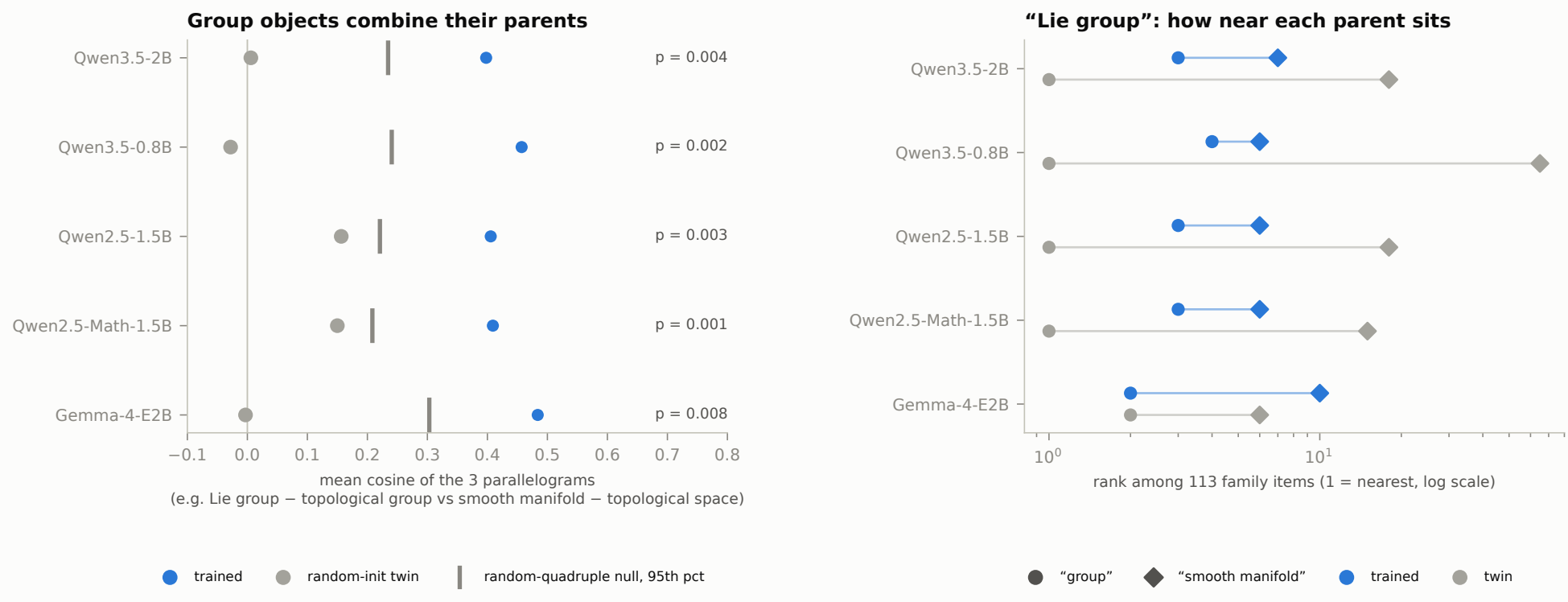
Group objects combine their parents
Qwen3.5-2B
Qwen3.5-0.8B
Qwen2.5-1.5B
Qwen2.5-Math-1.5B
Gemma-4-E2B
p = 0.004
p = 0.002
p = 0.003
p = 0.001
p = 0.008
−0.1
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
mean cosine of the 3 parallelograms
(e.g. Lie group − topological group vs smooth manifold − topological space)
trained
random-init twin
random-quadruple null, 95th pct
“Lie group”: how near each parent sits
Qwen3.5-2B
Qwen3.5-0.8B
Qwen2.5-1.5B
Qwen2.5-Math-1.5B
Gemma-4-E2B
10^0
10^1
rank among 113 family items (1 = nearest, log scale)
“group”
“smooth manifold”
trained
twin